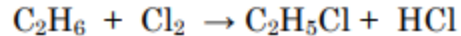
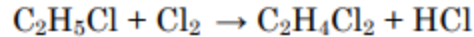


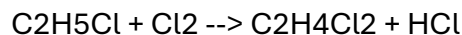
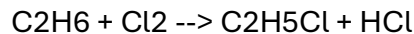
(1) Ethane is chlorinated in a continuous reactor:



Some of the product monochloroethane is further chlorinated in an undesired side reaction:



- (a) Suppose your objective is to maximize selectivity of the monochloroethane production relative to the dichloroethane production. Would you design the reactor for a high or low conversion of ethane? Explain your answer. What additional processing steps would almost certainly be necessary to make the process economical?
- (b) Draw and label a flowchart, assuming that the reactor feed contains only the two reactants. Use a degree-of-freedom analysis based on atomic species balances to determine how many process variable values must be specified for you to be able to calculate the remainder.
- (c) The reactor is designed to yield a 15% conversion of ethane and a selectivity of 14 moles of $\text{C}_2\text{H}_5\text{Cl}$ for every mole of $\text{C}_2\text{H}_4\text{Cl}_2$. Assume all of the chlorine gas is reacted. Calculate the feed ratio (moles Cl_2 / moles C_2H_6) and the fractional yield of monochloroethane.
- (d) Suppose the reactor is built and started up and the conversion is only 14%. Chromatographic analysis shows that there is no Cl_2 in the product, but another species of molecular weight higher than that of dichloroethane is present. Offer a likely explanation of the results.



a. Reactor conversion for best selectivity

The undesired reaction consumes the desired product

If ethane conversion is high, more $\text{C}_2\text{H}_5\text{Cl}$ is present in the reactor

This increases its chance to react further to $\text{C}_2\text{H}_4\text{Cl}_2$

So to maximize selectivity to monochloroethane we want low conversion per pass

Because low conversion $\rightarrow$ low concentration of intermediate product $\rightarrow$ less secondary chlorination

This is a series reaction selectivity problem

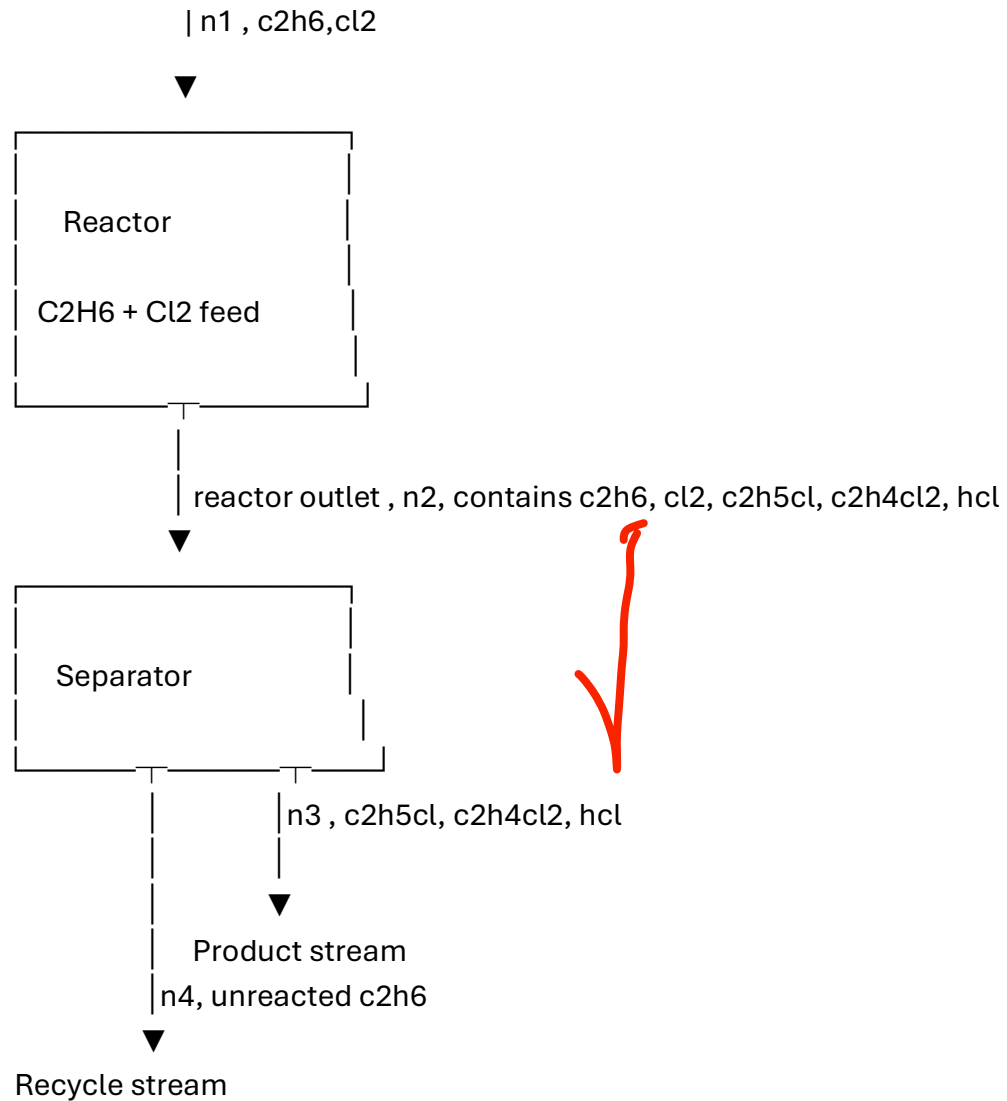
Additional processing required

Low conversion means most ethane exits unreacted so the plant must include the recycle of unreacted ethane

Additional separation steps likely required:

1. Product separation column
2. Recycle loop for ethane
3. Possibly HCl removal

b.



Dof analysis

Species involved

1. C2h6
2. Cl2
3. C2h5cl
4. C2h4cl2
5. Hcl

Atomic species

C

H

Cl

Independent balances:

3 atomic balances

Unknown outlet flowrates:

5 species

Reaction information adds:

2 independent reactions

So equations available:

3 (atom balances) + 2 (reaction relations) = 5

Unknowns: 5

DOF = $f - 5 = 0$

The system is exactly specified

~~4~~ - 3

c. Feed ratio and fractional yield

Ethane conversion

$X_{C_2H_6} = 0.15$

Selectivity

$S = C_2H_5Cl / C_2H_4Cl_2 = 14$

All Cl_2 consumed

Choosing a basis

100 mol C_2H_6

Conversion = 15%

C_2H_6 reacted = 15

Reaction 1 extent ξ_1

Reaction 2 extent = ξ_2

Reaction 1 consumes ethane:

$\xi_1 = 15$

$S = C_2H_5Cl / C_2H_4Cl$

Products formed:

Reaction 1 $\rightarrow C_2H_5Cl = \xi_1$

Reaction 2 consumes C_2H_5Cl and makes $C_2H_4Cl_2$

Net monochloroethane:

$C_2H_5Cl = \xi_1 - \xi_2$

Dichloroethane: $C_2H_4Cl_2 = \xi_2$

So

$14 = (\xi_1 - \xi_2) / \xi_2$

Substitute $\xi_1 = 15$

$14 = (15 - \xi_2) / (\xi_2)$

$$14x_2 = 15 - x_2$$

$$15x_2 = 15$$

$$x_2 = 1$$

Product amounts

$$C_2H_5Cl = 15 - 1 = 14$$

$$C_2H_4Cl_2 = 1$$

Cl₂ consumption reaction 1:

1 Cl₂ per reaction

Reaction 2:

1 Cl₂ per reaction

$$Cl_2 = 15 + 1 = 16$$

Since all Cl₂ reacts:

$$\text{Feed } Cl_2 = 16 \text{ mol}$$

Feed ratio

$$Cl_2/C_2H_6 = 16/100 = 0.16$$

Fractional yield of monochloroethane

$$Y = \text{moles desired product} / \text{moles ethane reacted}$$

$$Y = 14/15 = 0.933$$

$$Y = 93.3\%$$

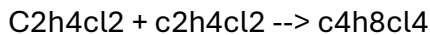
↓
X - 2 pts

- d. Conversion drops to 14% but no Cl₂ detected

No Cl₂, species with MW = 2 * dichloroethane

Further chlorination/polymerization reactions occurred

Example:

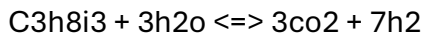


These heavier compounds consume Cl₂, reduce apparent ethane conversion, and appear as species with double molecular weight

↓

20/25

4.40



Given

1.2 atm

200 C

5:1 feed ratio H₂O : C₃H₈O₃

150 g/h glycerine feed

0.54 exit y_{H₂}

55 atm⁶ K_a

C₃H₈O₃ = 92 g/mol

$$N_{gly,in} = 150/92$$

$$N_{gly,in} = 1.63 \text{ mol/h}$$

Water feed (5:1 ratio)

$$N_{H_2O,in} = 5(1.63) = 8.15 \text{ mol/h}$$

X = fractional conversion of glycerine

species	initial	change	exit
glycerine	1.63	-1.63x	1.63(1-x)
water	8.15	-4.89x	8.15-4.89x
co2	0	4.89x	4.89x
h2	0	11.41x	11.41x

Total moles out

$$N_T = 1.63(1-x) + 8.15 - 4.89x + 4.89x + 11.41x$$

$$N_T = 9.78 + 6.52x$$

$$Y_{H_2} = (11.41x)/(9.78 + 6.52x)$$

$$Y_{H_2} = 0.54$$

$$0.54(9.78 + 6.52x) = 11.41x$$

$$5.281 + 3.52x = 11.41x$$

$$5.281 = 7.89x$$

$$X = 0.67$$

67% glycerine conversion

Is the reactor at equilibrium?

$$K_a = (p_C^3 \cdot p_H^7)/(p_G \cdot p_W^3)$$

$$P_i = y_i p$$

$$K_a = p^{(6)} \cdot (y_C^3 \cdot y_H^7)/(y_G \cdot y_W^3)$$

Using x = 0.67

Total moles

$$N_t = 9.78 + 6.52(0.67) = 14.15$$

species	mol
glycerine	0.54
water	4.87
co2	3.28
h2	7.64

Mole fractions

species	y
h2	0.54
co2	0.232
h2o	0.344
glycerine	0.038

$$Q = p^{(6)} * (y_c^3 * y_h^7) / (y_g * y_w^3)$$

$$P^6 = 1.2^6 = 2.99$$

$$Q = 2.99 * ((0.232)^3 * (0.54)^7) / ((0.038) * (0.344)^3)$$

$$Q = .53$$

$$K = 55$$

$$Q < K$$

The reactor is far from equilibrium

To improve conversion, increase the temperature, and remove h2 product (shifts the equilibrium right)

$$55 = (1.2)^6 * ((y_c)^3 * (y_h)^7) / ((y_g) * (y_w)^3)$$

Solving again gets

$$X_{eq} = 0.99$$

Equilibrium composition

$$N_t = 9.78 + 6.52(0.99) = 16.23$$

$$N_{h2} = 11.41(0.99) = 11.30$$

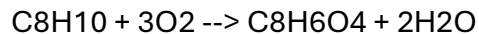
$$Y_{h2} = 0.70$$

X ~ 0.99

10/25

problem

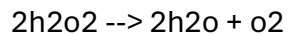
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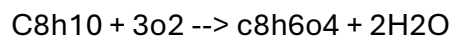
p-xylene C_8H_{10}

TA $C_8H_6O_4$

Hydrogen peroxide decomposes



Oxygen then oxidizes p-xylene



Basis

100 kg/day TA

Molecular weights

TA – 166

p-xylene – 106

O₂ – 32

H₂O – 34

$N_{ta} = 100000/166$

$N_{ta} = 602 \text{ mol/day}$

Conventional process calculations

p-xylene required 1:1

$N_{xylene} = 602 \text{ mol}$

Mass

$602(106) = 63.8 \text{ kg}$

Oxygen required

$3(602) = 1806 \text{ mol}$

Air feed

Air contains 21% oxygen

$N_{air} = 1806/0.21$

$N_{air} = 8600 \text{ mol}$

With 20% excess air

$1.2(8600) = 10320 \text{ mol}$

Acetic acid solvent

p-xylene is 10 wt%

$M_{solution} = 63.8/0.10 = 638 \text{ kg}$

Acetic acid

574 kg

10% oxidation of acetic acid

Loss

$$0.10(574) = 57.4 \text{ kg}$$

Conventional process atom economy

Atom economy AE

$$AE = \frac{MW_{\text{product}}}{\sum MW_{\text{reactants}}}$$

$$\text{Reactants } 106 + 96 = 202$$

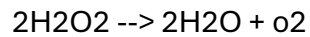
$$AE = 166/202$$

$$AE = 82\%$$



Acetic acid oxidation wastes material, lowering overall efficiency

Hydrogen peroxide produces oxygen



Needed oxygen

$$1806 \text{ mol}$$

Hydrogen peroxide required

$$2(1806)$$

$$3612 \text{ mol}$$

$$\text{Mass } 3612(34)$$

$$122.8 \text{ kg}$$

p-xylene required is the same as before 63.8 kg

New process AE

$$\text{p-xylene} - 106\text{g}$$

$$\text{H}_2\text{O}_2 (2 \text{ mol}) - 68\text{g}$$

$$\text{Total } 174 \text{ g}$$

$$AE = 166/174 = 95\%$$

X - 3 lbs

p-xylene + acetic acid $\rightarrow$ oxidation reactor $\rightarrow$ TA crystallization $\rightarrow$ TA product

|

\

--> solvent recycle + waste CO₂/H₂O

Air provides oxygen

p-xylene + H₂O₂ --> supercritical water reactor --> TA separation --> water byproduct

Quantity	Conventional	New process
p-xylene	63.8 kg	63.8 kg
oxidant	air	122.8 kg H ₂ O ₂
solvent	574 kg acetic acid	none
solvent loss	57 kg/day	none
atom economy	82%	95%

New process pros and cons:

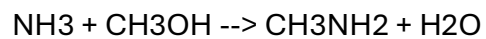
Higher atom economy, no solvent oxidation, fewer separation steps, and higher selectivity, but higher hydrogen peroxide cost, supercritical reactor cost, and peroxide handling safety

I would use this new process, because of its fewer steps, and higher atom economy

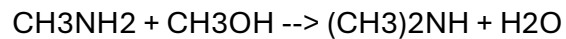
20/25 — 2.05

Problem 4.50

Reaction 1 methylamine



R2 dimethylamine formation



Basis and outlet composition

Feed

100 gmol NH₃/h, and 100 gmol CH₃OH/h

Reactor effluent mole fractions

Nh3: 20 mol%

Ch3oh: 9.4 mol%

Ch3nh2: 18.6 mol%

(ch3)2nh: 10.9 mol%

H2O: remainder

So water is:

$$100 - 20 - 9.4 - 18.6 - 10.9 = 41.1\%$$

$$Y_{nh3} = 0.2, y_{meoh} = 0.094, y_{mma} = 0.186, y_{dma} = 0.109, y_{h2o} = 0.411$$

Let the total outlet flowrate be F gmol/h

Then the outlet component flowrates are

$$N_{nh3} = 0.200F$$

$$N_{meoh} = 0.094F$$

$$N_{mma} = 0.186F$$

$$N_{dma} = 0.109F$$

$$N_{h2o} = 0.411F$$

Carbon balance

Carbon in

Only methanol contains carbon in the feed, 1 C per methanol so $C_{in} = 100$ gmol C/h

Carbon out

Methanol: $0.094 F$ mol, 1 C each

Methylamine: $0.186 F$ mol, 1 C each

Dimethylamine: $0.109F$ mol, 2 C each

$$C_{out} = 0.094 F + 0.186 F + 2(0.109F)$$

$$C_{out} = 0.498 F$$

Set in = out

$$100 = 0.498 F$$

$$F = 200.8 \text{ gmol/h}$$

Outlet flowrates using $F = 200.8$

$$N_{nh3} = 0.200(200.8) = 40.16$$

$$N_{meoh} = 0.094(200.8) = 18.88$$

$$N_{mma} = 0.186(200.8) = 37.35$$

$$N_{dma} = 0.109(200.8) = 21.89$$

$$N_{h2o} = 0.411(200.8) = 82.53$$

Outlet stream:

$$N_{h3} \quad 40.16 \text{ gmol/h}$$

$$Ch3oh \quad 18.88 \text{ gmol/h}$$

$$Ch3nh2 \quad 37.35 \text{ gmol/h}$$

$$(ch3)2nh \quad 21.89 \text{ gmol/h}$$

$$H2o \quad 82.53 \text{ gmol/h}$$

Nitrogen balance

Nitrogen in (only nh3)

$$N_{in} = 100$$

Nitrogen out

$$N_{h3}: 40.16$$

$$\text{Methylamine: } 37.35$$

$$\text{Dimethylamine: } 21.89$$

$$N_{out} = 40.16 + 37.35 + 21.89 = 99.40$$

This is very close to 100, error is most likely from rounding

Oxygen balance

Oxygen in

Only methanol contains oxygen in the feed

$$O_{in} = 100$$

Oxygen out

Methanol: 18.88

Water 82.53

$$O_{out} = 18.88 + 82.53 = 101.41$$

Again, very close.

Hydrogen balance

Hydrogen in

$$NH_3: 100 * 3 = 300$$

$$Methanol : 100 * 4 = 400$$

$$H_{in} = 700$$

Hydrogen out

$$NH_3: 40.16 * 3 = 120.48$$

$$Methanol: 18.88 * 4 = 75.52$$

$$Methylamine: 37.35 * 5 = 186.75$$

$$Dimethylamine 21.89 * 7 = 153.23$$

$$Water 82.53 * 2 = 165.06$$

$$H_{out} = 120.48 + 75.52 + 186.75 + 153.23 + 165.06 = 701.04$$

This analysis is reasonable, the errors tend to be quite small and most likely due to rounding

Reaction rates/ extents

R1 = rate of reaction 1 in gmol/h

R2 = rate of reaction 2 in gmol/h

Steady-flow reactor, and no products are in the feed

Dimethylamine balance

Only reaction 2 makes dimethylamine

$$R_2 = n_{DMA,out}$$

$$R_2 = 21.89 \text{ gmol/h}$$

Methylamine balance

Reaction 1 makes methylamine; reaction 2 consumes it

$$N_{\text{mma,out}} = r_1 - r_2$$

So

$$R_1 = n_{\text{mma,out}} + r_2$$

$$R_1 = 37.35 + 21.89 = 59.24$$

$$R_1 = 59.2 \text{ gmol/h}$$

Fractional conversion of methanol

Methanol feed:

$$N_{\text{meoh,in}} = 100$$

Methanol out

$$N_{\text{meoh,out}} = 18.88$$

Methanol reacted:

$$100 - 18.88 = 81.12$$

Fractional conversion:

$$X_{\text{meoh}} = (100 - 18.88) / 100 = 0.811$$

$$X_{\text{meoh}} = 0.811$$

So methanol conversion is

$$81.1\%$$

Yield of methylamine based on methanol

Yield of desired product based on reactant = net desired product formed / reactant fed

Net methylamine formed

$$N_{\text{mma,out}} = 37.35$$

Methanol fed:

$$100$$

Therefore

$$Y_{\text{mma/meoh}} = 37.35/100 = 0.3735$$

Methylamine yield based on methanol feed is 37.35%

Selectivity for methylamine based on methanol

$S_{\text{mma/meoh}} = \text{net desired product formed} / \text{reactant consumed}$

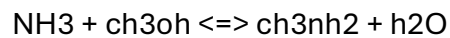
$$S_{\text{mma/meoh}} = 37.35/81.12 = 0.4605$$

46.1%

Is the reactor effluent at equilibrium

$$K_1 = 4$$

$$K_2 = 2.5$$

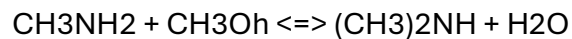


$$Q_1 = ((y_{\text{mma}})(y_{\text{H}_2\text{O}})) / ((y_{\text{NH}_3})(y_{\text{meoh}}))$$

$$Q_1 = ((0.186)(0.411)) / ((0.200)(0.094))$$

$$Q_1 = 4.07$$

$K_1 = 4.0$, so this is basically at equilibrium , more or less



$$Q_2 = ((y_{\text{dma}})(y_{\text{H}_2\text{O}})) / ((y_{\text{mma}})(y_{\text{meoh}}))$$

$$Q_2 = ((0.109)(0.411)) / ((0.186)(0.094))$$

$$Q_2 = 2.56$$

$$K_2 = 2.5$$

Basically at equilibrium again

So yeah, the reactor is effluent at equilibrium

Optimization of feed ratio $\text{NH}_3 : \text{CH}_3\text{OH}$

Basis of 1 mol methanol fed

R mol ammonia fed

So feed ratio is

$$R = \text{NH}_3 \text{ fed} / \text{CH}_3\text{OH fed}$$

ξ_1 is for reaction 1 and ξ_2 is for reaction 2

Outlet mole numbers are

$$N_{\text{NH}_3} = R - \xi_1$$

$$N_{\text{MeOH}} = 1 - \xi_1 - \xi_2$$

$$N_{\text{DMA}} = \xi_1 - \xi_2$$

$$N_{\text{DMA}} = \xi_2$$

$$N_{\text{H}_2\text{O}} = \xi_1 + \xi_2$$



species	feed	change	outlet
nh3	R	$-\xi_1$	$R - \xi_1$
ch3oh	1	$-\xi_1 - \xi_2$	$1 - \xi_1 - \xi_2$
ch3nh2	0	$\xi_1 - \xi_2$	$\xi_1 - \xi_2$
(ch3)2nh	0	ξ_2	ξ_2
h2o	0	$\xi_1 + \xi_2$	$\xi_1 + \xi_2$

Total moles don't change

$$N_t = R + 1$$

Equilibrium equations

R1

$$K_1 = 4 = \frac{(N_{\text{DMA}})(N_{\text{H}_2\text{O}})}{(N_{\text{NH}_3})(N_{\text{MeOH}})}$$

$$4 = \frac{(\xi_1 - \xi_2)(\xi_1 + \xi_2)}{(R - \xi_1)(1 - \xi_1 - \xi_2)}$$

R2

$$K_2 = 2.5 = \frac{(N_{\text{DMA}})(N_{\text{H}_2\text{O}})}{(N_{\text{DMA}})(N_{\text{MeOH}})}$$

$$2.5 = \frac{\xi_2(\xi_1 + \xi_2)}{(N_{\text{DMA}})(N_{\text{MeOH}})}$$

Maximum methylamine concentration in outlet

$$Y_{mma} = (x_{i1} - x_{i2}) / (R + 1)$$

$$2.5(x_{i1} - x_{i2})(1 - x_{i1} - x_{i2}) = x_{i2}(x_{i1} + x_{i2})$$

Solving with R1 we get

$$x_{i1} = 0.50 \text{ and } x_{i2} = 0.21 \text{ (used solver on excel) when } R = 1.44$$

$$R = 1.44$$

$$\text{NH}_3 : \text{CH}_3\text{OH} = 1.44 : 1$$

$$N_t = R + 1 = 2.44$$

Methylamine produced

$$N_{mma} = x_{i1} - x_{i2}$$

$$= 0.50 - 0.21$$

$$= 0.29$$

$$Y_{mma} = 0.29 / 2.44$$

$$Y_{mma} = 0.198$$

Basis

1 mol CH₃OH fed, R mol NH₃ fed where

$$R = \text{NH}_3 \text{ fed} / \text{CH}_3\text{OH fed}$$

$$N_{mma} / n_{dma} = (x_{i1} - x_{i2}) / x_{i2}$$

Rearranging the second equilibrium equation to

$$(x_{i1} - x_{i2}) / x_{i2} = (x_{i1} + x_{i2}) / (2.5(1 - x_{i1} - x_{i2}))$$

If $1 - x_{i1} - x_{i2}$ gets smaller the ratio gets bigger

$1 - x_{i1} - x_{i2}$ is the methanol left in the reactor, so the ratio gets larger when methanol becomes smaller

When R becomes very large, NH_3 is in excess, R1 is driven forward, r2 is suppressed relative to R1 because the system tends to keep more MMA and less DMA, methanol can be almost completely consumed

Basically $x = 1 - x_{i1} - x_{i2}$

Where x is the methanol remaining

When R is very large, then x gets closer to zero

$$2.5 = \frac{x_{i2}(x_{i1} + x_{i2})}{(x_{i1} - x_{i2})x}$$

$$\text{So } \frac{x_{i1} - x_{i2}}{x_{i2}} = \frac{x_{i1} + x_{i2}}{(2.5x)}$$

As x goes to zero, then $(x_{i1} - x_{i2})/x_{i2} \rightarrow \text{infinity}$

Because of this, the ratio MMA/DMA increases infinitely as $R \rightarrow \text{infinity}$

So to maximize the methylamine to dimethylamine ratio, use a very large excess of ammonia

Maximizing yield of methylamine based on ammonia

R is based on NH_3 and net methylamine produced is $x_{i1} - x_{i2}$, $y_{\text{MMA}}/\text{NH}_3 = (x_{i1} - x_{i2})/R$

Ok so if we set $a = x_{i1} + x_{i2}$

$$b = x_{i1} - x_{i2}$$

$$\text{Then } x_{i1} = (a+b)/2, x_{i2} = (a-b)/2$$

Then the outlet moles become $n_{\text{H}_2\text{O}} = a$, $n_{\text{MMA}} = b$

$$n_{\text{CH}_3\text{OH}} = 1 - a$$

$$n_{\text{NH}_3} = R - (a+b)/2$$

Rewriting equation 2 using a and b

$$2.5 = \frac{((a-b)/2) \cdot a}{(b(1-a))}$$

$$\text{Solving for } b = \frac{a^2}{(5-4a)}$$

Rewriting equation 1

With a and b

$$4 = \frac{(ba)}{(R - ((a+b)/2))(1-a)}$$



$$2(1-a)(2R-a-b) = ab$$

Substituting $b = a^2/(5-4a)$ into the last equation and simplifying

$$R = (a(7a^2 - 16a + 10))/((4(4a^2 - 9a + 5)))$$

$$Y_{mma}/nh^3 = b/R$$

Taking

$$R = (a(7a^2 - 16a + 10))/((4(4a^2 - 9a + 5)))$$

Move it all to one side

$$7a^3 - (16 + 16R)a^2 + (10 + 36R)a - 20R = 0$$

For each R solve this for some root $0 < a < 1$

Putting a few examples into a table

R	A = xi1 + xi2	B = xi1 - xi2	xi1	xi2	Ymma/nh3 = b/R
0.50	0.6531	0.1787	0.4159	0.2372	0.357
0.75	0.7570	0.2905	0.5238	0.2332	0.387
1.00	0.8119	0.3762	0.5940	0.2179	0.376
1.50	0.8687	0.4948	0.6817	0.1870	0.330
2.00	0.8982	0.5733	0.7357	0.1625	0.287

The maximum y_{mma}/nh^3 is closer to $R = 0.75$

To find the exact root, I would need to take b/r and solve for it in terms of a

$$Y(a) = (4a(1-a))/(7a^2 - 16a + 10)$$

(this is a simplified version of b/r in terms of a . $((a^2)/(5-4a)) / ((a(7a^2 - 16a + 10))/(4(4a^2 - 9a + 5)))$)

The derivative of this is (set equal to zero) $9a^2 - 20a + 10 = 0$

Solving for a using quadratic

$$A = (10 \pm \sqrt{10})/9$$

$10 - 10\sqrt{10} / 9$ is the reasonable root, which is 0.7597

Using $a = 0.7597$ in the equation for R in terms of a gets $R = 0.7597$, meaning the optimum ammonia: methanol feed ratio is 0.76:1

And the maximum yield is $y_{\max} = b/R = 0.3874$

So $y_{\text{mma}/\text{nh}_3, \max} = 0.387$

And finally, maximizing yield of methylamine based on methanol

$$Y_{\text{mma}/\text{meoh}} = x_1 - x_2$$

We follow basically the same steps as c, up to the point where we get R in terms of a . before we differentiated and set equal to zero for roots, this time we don't

$$dy/da \text{ simplifies to } (2a(5-2a))/(5-4a)^2$$

With the range $0 < a < 1$, $dy/da > 0$

So the yield increases with a .

As a gets closer to 1, almost all methanol is consumed. The numerator gets closer to $1(7-16+10) = 1$ and the denominator gets closer to $4(4-9+5) = 0$

So as a approaches 1 r approaches infinity

$$Y_{\text{mma}/\text{meoh}} = a^2/(5-4a)$$

$$Y_{\max} = \lim_{a \rightarrow 1} a^2/(5-4a)$$

$$Y_{\max} = 1/1 = 1 \text{ (substituting 1 into the limit)}$$

So the maximum possible yield based on methanol is 100%

If ammonia is in large excess, reaction 1 is strongly favored, methanol is driven almost completely into methylamine formation

Dimethylamine formation becomes negligible compared with methylamine formation, so net methylamine produced approaches 1 mol per 1 mol methanol fed

Therefore large ammonia excess gives nearly complete conversion of methanol to methylamine

This problem never ends